

TASK — BODY LANDMARK DETECTION SYSTEM

AI/ML Internship — Day 7

Name: Praneetha AS

Role: AI/ML Intern

Project Folder: Raritone

Module: Human Body Detection System

1. Introduction

The objective of this task was to develop and test a Human Body Landmark Detection System using Artificial Intelligence and Computer Vision techniques.

The project focused on:

- detecting human body landmarks
- skeletal visualization
- pose estimation
- body tracking
- landmark analysis

The system was implemented using:

- Python
- OpenCV
- MediaPipe

The AI model detects body joints and generates real-time skeletal tracking output from images.

2. Objective

The main objective of this module was to:

- detect human body landmarks
 - analyse body posture
 - generate skeletal visualization
 - improve pose tracking accuracy
 - study AI pose estimation workflows
-

3. Technologies Used

Technology	Purpose
Python	Programming
OpenCV	Image processing
MediaPipe	Pose estimation
Artificial Intelligence	Landmark analysis
Computer Vision	Body tracking

4. Human Body Landmark Detection

The AI system detects several body landmarks including:

- nose
- shoulders
- elbows
- wrists
- hips
- knees
- ankles

These landmarks help the AI system understand body posture and movement.

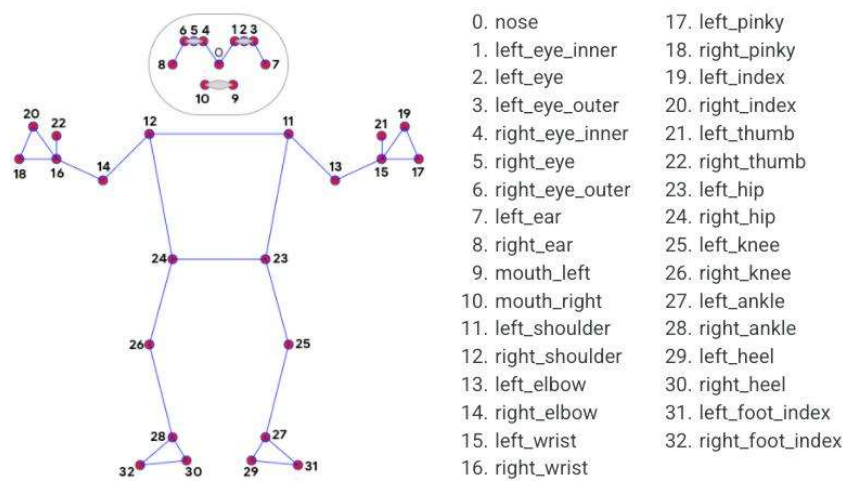


Figure 1 — Human Landmark Detection Output

5. Workflow of Body Detection System

The AI workflow consists of multiple stages.

Stage 1 — Image Input

The system receives:

- body images
 - webcam frames
 - pose samples
-

Stage 2 — Human Detection

The AI system identifies:

- human body
 - body orientation
 - posture structure
-

Stage 3 — Landmark Extraction

MediaPipe detects:

- shoulder points
 - elbow points
 - wrist points
 - hip points
 - knee points
-

Stage 4 — Skeletal Visualization

The AI system connects body landmarks and generates:

- skeletal lines
 - body structure visualization
 - landmark tracking
-

Figure 2 — Skeletal Tracking Visualization

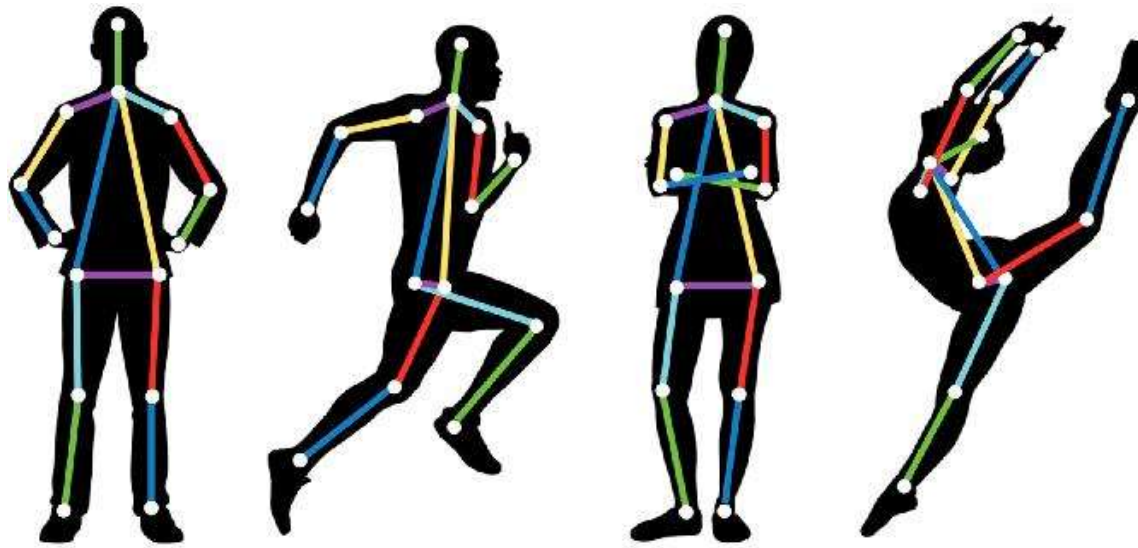


Figure 2 — Skeletal Tracking Visualization

6. Landmark Detection Demo

The landmark detection demo successfully:

- detected body posture
- generated body landmarks
- visualized skeletal connections
- tracked human body movement

The AI system processed:

- standing poses
- side poses
- arm movement positions
- different body orientations

7. Python Implementation

```
import cv2
```

```
import mediapipe as mp
```

```
# Initialize MediaPipe Pose
```

```
mp_pose = mp.solutions.pose
```

```
mp_draw = mp.solutions.drawing_utils

pose = mp_pose.Pose(
    static_image_mode=True,
    model_complexity=2,
    min_detection_confidence=0.7
)

# Load image
image = cv2.imread(
    r"P:\AI_ML_Internship\Projects\Body_Landmark_Detection\walking.jpg"
)

# Check image
if image is None:
    print("Image not found")
    exit()

print("Image loaded successfully")

# Resize image
image = cv2.resize(image, (720, 720))

# Convert BGR to RGB
rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Process image
results = pose.process(rgb)

# Draw landmarks
```

```
if results.pose_landmarks:

    mp_draw.draw_landmarks(
        image,
        results.pose_landmarks,
        mp_pose.POSE_CONNECTIONS
    )

    print("Landmarks detected successfully")

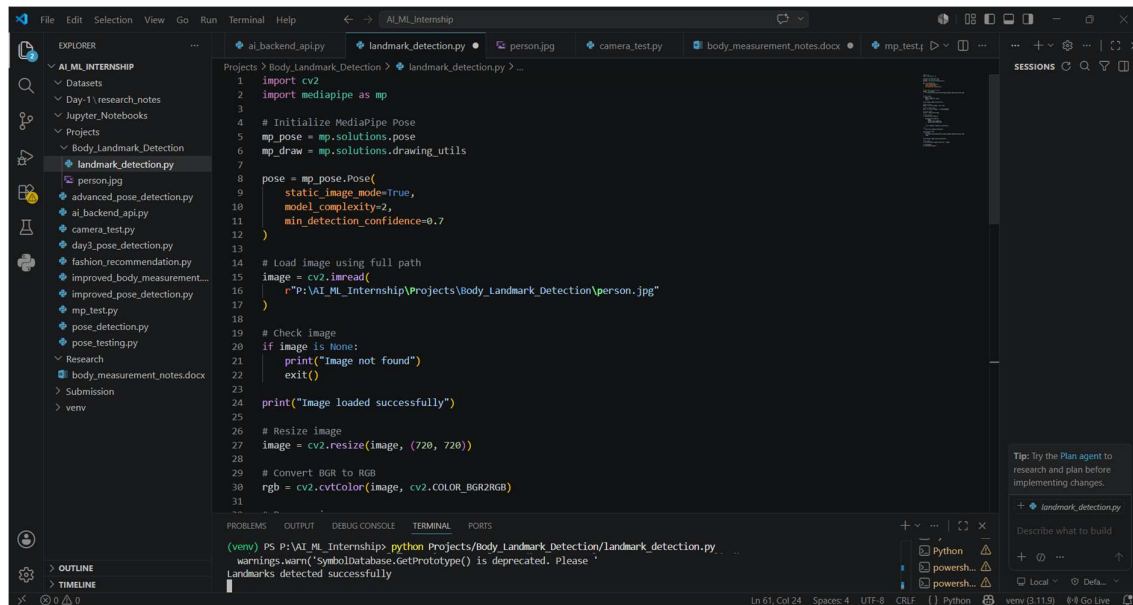
else:
    print("No landmarks detected")

# Save output image
cv2.imwrite(
    r"P:\AI_ML_Internship\Projects\Body_Landmark_Detection\output.jpg",
    image
)

print("Output image saved successfully")

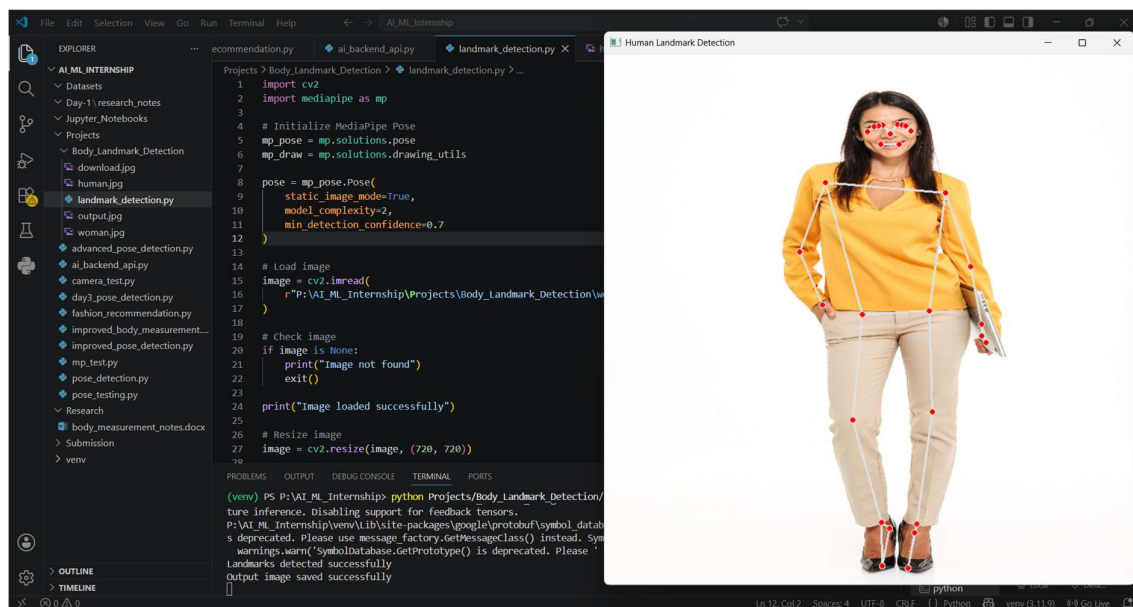
# Show image
cv2.imshow("Human Landmark Detection", image)

cv2.waitKey(0)
cv2.destroyAllWindows()
```



```
1 import cv2
2 import mediapipe as mp
3
4 # Initialize MediaPipe Pose
5 mp_pose = mp.solutions.pose
6 mp_draw = mp.solutions.drawing_utils
7
8 pose = mp_pose.Pose(
9     static_image_mode=True,
10    model_complexity=2,
11    min_detection_confidence=0.7
12)
13
14 # Load image using full path
15 image = cv2.imread(
16     r"P:\AI_ML_Internship\Projects\Body_Landmark_Detection\person.jpg"
17)
18
19 # Check image
20 if image is None:
21     print("Image not found")
22     exit()
23
24 print("Image loaded successfully")
25
26 # Resize image
27 image = cv2.resize(image, (720, 720))
28
29 # Convert BGR to RGB
30 rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
31
```

(venv) PS P:\AI_ML_Internship> python Projects\Body_Landmark_Detection\Landmark_detection.py ...
warnings.warn("SymbolDatabase.GetPrototype() is deprecated. Please ...
Landmarks detected successfully



8. Measurement Logic

The AI system estimates body measurements using landmark distances.

The system calculates:

- shoulder width
- body proportions
- arm length
- torso alignment

using:

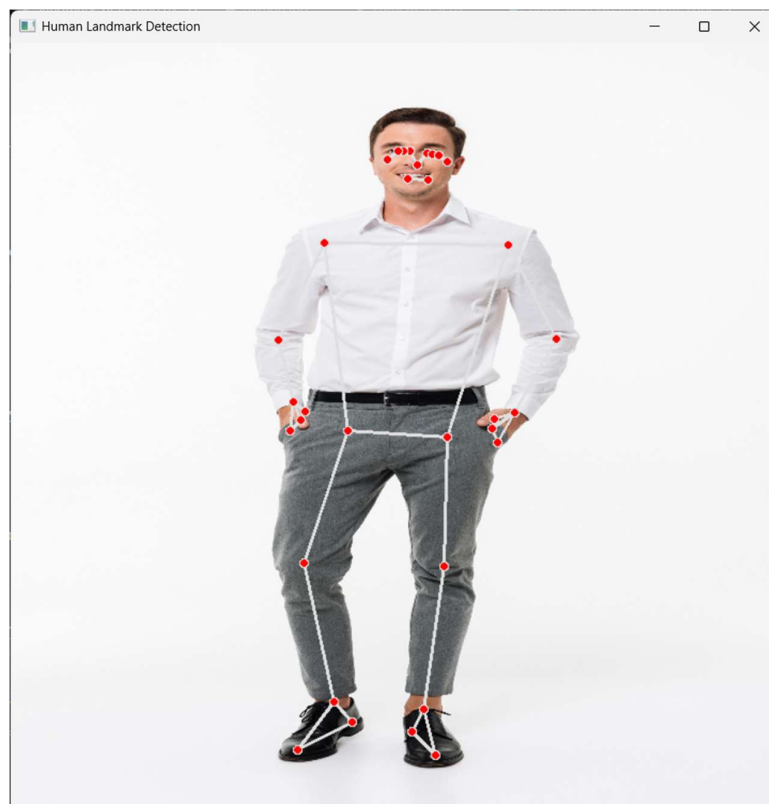
- landmark coordinates
- pixel distance calculations
- body proportion estimation

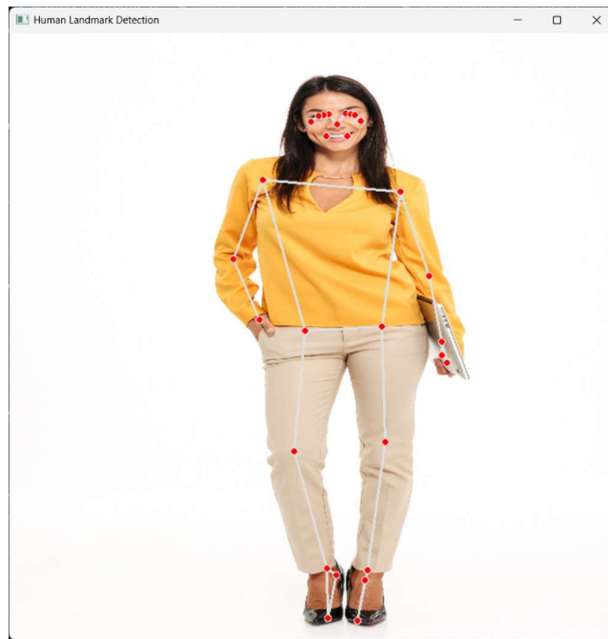
The measurement logic helps improve:

- virtual fitting systems
- body analysis
- pose estimation accuracy

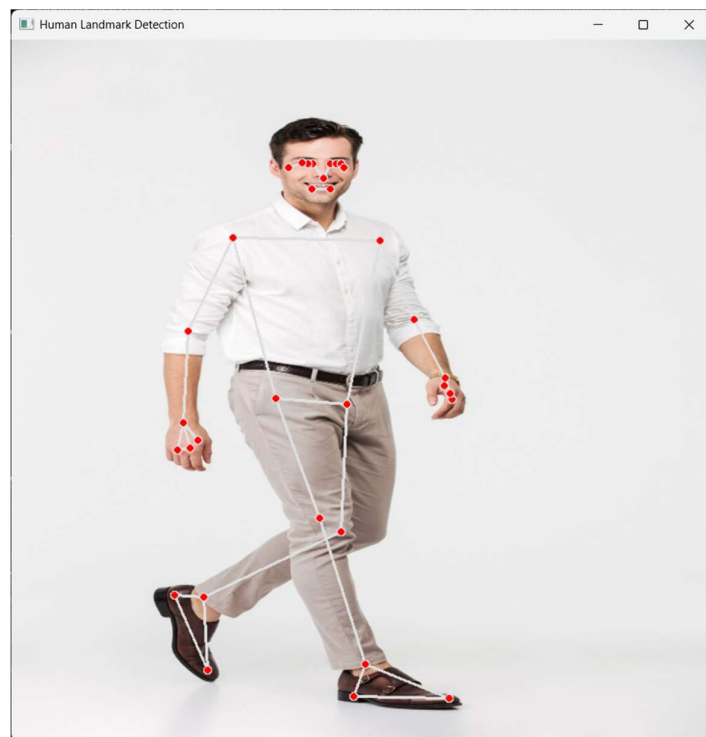
9. Testing Observations

Observation	Result
Standing pose	Accurate detection
Side body pose	Minor landmark variation
Arm movement	Smooth tracking
Good lighting	High accuracy
Poor lighting	Reduced landmark stability





Standing Position



Detection while Walking

Figure 3 — Body Landmark Testing Output

10. Challenges Faced

Challenge	Impact
Poor lighting	Reduced landmark accuracy
Fast movement	Tracking instability
Complex background	Detection variation
Side body angle	Minor landmark errors

11. Learning Outcomes

During this task, the following concepts were understood:

- human pose estimation
 - body landmark tracking
 - skeletal visualization
 - AI pose workflow
 - computer vision processing
 - landmark analysis systems
-

12. Conclusion

The Human Body Landmark Detection System successfully detected and tracked human body landmarks using Artificial Intelligence and Computer Vision technologies.

The task improved understanding of:

- AI pose estimation
- skeletal tracking
- landmark detection systems
- body measurement logic
- AI body analysis
- computer vision workflows

The system demonstrated how AI-based pose estimation technologies are used in:

- virtual try-on systems
- healthcare applications
- fitness tracking
- AI fashion technology
- body analysis systems